Topic 1: Basic Linux Commands

1. **Pwd ~** Display the current working directory

```
sanjana3027@DESKTOP-QJF5AGU:~$ pwd
/home/sanjana3027
```

2. **Clear ~ clear the terminal screen**

```
sanjana3027@DESKTOP-QJF5AGU:~$
```

3. **History ~ displays previously executed commands**

```
48  ./FourthProgram
49  ls
50  cat Hello.c
51  nano Hello.c
52  pwd
53  clear
```

4. **Whoami ~ display the current logged-in username**

```
sanjana3027@DESKTOP-QJF5AGU:~$ whoami
sanjana3027
```

5. **Date ~ display the current system date and time**

```
sanjana3027@DESKTOP-QJF5AGU:~$ date
Tue Aug 11 04:05:51 UTC 2026
```

6. **Hostname ~ Display the system hostname**

```
sanjana3027@DESKTOP-QJF5AGU:~$ hostname
DESKTOP-QJF5AGU
```

7. **Uname -a ~ Display operating system operation**

```
sanjana3027@DESKTOP-QJF5AGU:~$ uname -a
Linux DESKTOP-QJF5AGU 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_6
inux
```

8. **Exit ~ exit the terminal**

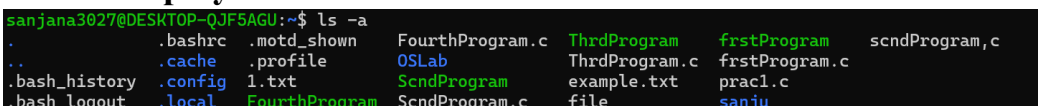
```
sanjana3027@DESKTOP-QJF5AGU:~$ exit
```

TOPIC 2 ~ DIRECTORY MANAGEMENT

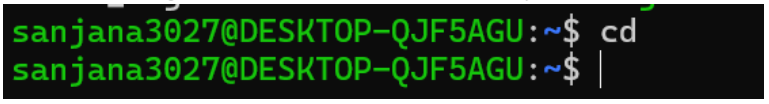
1. **Ls ~ list files and directories**



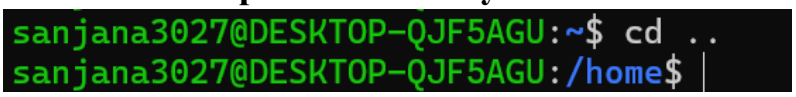
2. **Ls -a ~ display detailed file information**



3. **Cd ~ change the current directory**



4. **Cd ..~ move to parent directory**



5. **mkdir ~ create a new directory**



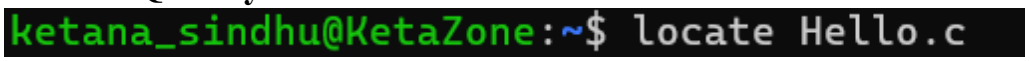
6. **rmdir ~ remove a directory**



7. **find FileName.c ~ search for files and directories**



8. **locate ~ Quickly locate files**



3.FILE MANAGEMENT

1.touch ~ Create new empty file

```
ketana_sindhu@KetaZone:~$ touch KL.c
```

2. cat ~ display file contents

```
ketana_sindhu@KetaZone:~$ cat Hello.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$
```

3. cp ~ copy files or directories

```
ketana_sindhu@KetaZone:~$ cp Hello.c FirstFile.c
ketana_sindhu@KetaZone:~$
```

4. mv ~ move or rename files

```
ketana_sindhu@KetaZone:~$ mv Hello.c hello.c
ketana_sindhu@KetaZone:~$
```

5. rm ~ remove file

```
ketana_sindhu@KetaZone:~$ rm hello.c
ketana_sindhu@KetaZone:~$
```

6. head ~ display first lines of a file

```
ketana_sindhu@KetaZone:~$ head FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$
```

7. tail ~ display last lines of the file

```
ketana_sindhu@KetaZone:~$ tail FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$
```

8.more ~ view a file page by page

```
ketana_sindhu@KetaZone:~$ more FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$
```

9.less ~ Scroll through a file

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
FirstFile.c (END)
```

10.wc ~ count lines, words and characters

```
ketana_sindhu@KetaZone:~$ wc FirstFile.c
 7  9 77 FirstFile.c
ketana_sindhu@KetaZone:~$
```

11.diff ~ compare two files

```
ketana_sindhu@KetaZone:~$ diff FirstFile.c PID.c
1a2
> #include<unistd.h>
5c6,19
<      printf("Hello Linux\n");
---
>      pid_t pid;
>
>      pid=fork();
>
>      if(pid==0)
>      {
>          printf("Child PID = %d\n",getpid());
>          printf("Parent PID = %d\n",getppid());
>      }
>      else
>      {
>          printf("Parent PID = %d\n",getpid());
>      }
>
ketana_sindhu@KetaZone:~$
```

12.sort ~ sort file contents

```
ketana_sindhu@KetaZone:~$ sort FirstFile.c

    printf("Hello Linux\n");
    return 0;
#include<stdio.h>
int main()
{
}
ketana_sindhu@KetaZone:~$
```

13.uniq ~ remove duplicate files

```
ketana_sindhu@KetaZone:~$ uniq FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$
```

14.grep ~ search for text inside a file

```
ketana_sindhu@KetaZone:~$ grep main FirstFile.c
int main()
ketana_sindhu@KetaZone:~$
```

4. TOPIC ~ FILE EDITING

1.nano ~ create or edit a file

```
ketana_sindhu@KetaZone:~$ nano Hello.c
ketana_sindhu@KetaZone:~$
```

2. vi ~ edit a file using vim editor

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

5. C PROGRAM COMPILATION AND EXECUTION

1. **gcc ~ compile a C program**

```
ketana_sindhu@KetaZone:~$ gcc FirstFile.c -o file
```

2. **. /program ~ Execute the compiled program**

```
ketana_sindhu@KetaZone:~$ gcc FirstFile.c -o file
ketana_sindhu@KetaZone:~$ ./file
Hello Linux
```

3. **which ~ Locate the executable of a command**

```
ketana_sindhu@KetaZone:~$ which gcc
/usr/bin/gcc
ketana_sindhu@KetaZone:~$
```

4. **whereis ~ Locate binary, source and manual pages**

```
ketana_sindhu@KetaZone:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gc
ketana_sindhu@KetaZone:~$
```

5. **man ~ display manual pages**



6.PROCESS MANAGEMENT

1. ps ~ display running processes

```
ketana_sindhu@KetaZone:~$ ps
    PID TTY          TIME CMD
    343 pts/0    00:00:00 bash
    467 pts/0    00:00:00 ps
ketana_sindhu@KetaZone:~$
```

2. ps -ef ~ display all running processes

```
ketana_sindhu@KetaZone:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 15:31 ?        00:00:00 /sbin/i
root           2       1  0 15:31 hvc0     00:00:00 /init
root           6       2  0 15:31 hvc0     00:00:00 plan9 -
root          46       1  0 15:31 ?        00:00:00 /usr/li
systemd+      82       1  0 15:31 ?        00:00:00 /usr/li
root          89       1  0 15:31 ?        00:00:00 /usr/li
root         144       1  0 15:31 ?        00:00:00 /bin/sh
root         151       1  0 15:31 ?        00:00:00 /usr/sb
message+     160       1  0 15:31 ?        00:00:00 @dbus-d
root         165       1  0 15:31 ?        00:00:00 /usr/bi
```

3. top ~ monitor running processes

```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMA
    1 root      20   0  24552  15572  11616 S   0.0   0.2   0:00.66 syste
    2 root      20   0   3180   2208   2072 S   0.0   0.0   0:00.01 init-
    5 root      20   0   3180   2048   1940 S   0.0   0.0   0:00.00 init
   46 root      19  -1  42200  16676  15416 S   0.0   0.2   0:00.11 syste
   82 systemd+  20   0  22540  14592  11972 S   0.0   0.2   0:00.05 syste
   89 root      20   0  35436  12248   9084 S   0.0   0.2   0:00.25 syste
  144 root      20   0   2888   1944   1820 S   0.0   0.0   0:00.07 chron
  151 root      20   0  44472  30952  28444 S   0.0   0.9   0:00.09 chron
  160 message+  20   0   8828   5392   4708 S   0.0   0.1   0:00.05 dbus-
  165 root      20   0  44092  29760  14616 S   0.0   0.4   0:00.24 netwo
  174 root      20   0  18432   9408   8168 S   0.0   0.1   0:00.12 syste
  177 root      20   0 1792300  15380  12548 S   0.0   0.2   0:00.12 wsl-p
  187 syslog    20   0  220548   5504   4364 S   0.0   0.1   0:00.07 rsysl
  255 _chrony   20   0  20420   6336   5564 S   0.0   0.1   0:00.02 chron
  263 root      20   0  123456  32616  18004 S   0.0   0.4   0:00.13 unatt
  277 _chrony   20   0  12092   2424   1784 S   0.0   0.0   0:00.00 chron
  300 root      20   0   5492   2768   2580 S   0.0   0.0   0:00.00 agett
  341 root      20   0   3184   1112    980 S   0.0   0.0   0:00.00 Sessi
  342 root      20   0   3200   1252   1108 S   0.0   0.0   0:00.00 Relay
  343 ketana_+  20   0   6268   5540   3864 S   0.0   0.1   0:00.05 bash
  344 root      20   0   8648   5624   4780 S   0.0   0.1   0:00.00 login
  386 ketana_+  20   0  22332  13508  10880 S   0.0   0.2   0:00.07 syste
  388 ketana_+  20   0  22908   4088   2088 S   0.0   0.1   0:00.00 (sd-p
  416 ketana_+  20   0   6136   5396   3848 S   0.0   0.1   0:00.01 bash
  499 ketana_+  20   0   8172   5992   3888 R   0.0   0.1   0:00.01 top
```

4. kill ~ terminate a process

```
ketana_sindhu@KetaZone:~$ kill 1234
-bash: kill: (1234) - No such process
```

5.kill -9 ~ Forcefully terminate a process

```
ketana_sindhu@KetaZone:~$ kill -9
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
ketana_sindhu@KetaZone:~$
```

7.JOB CONTROL

1. jobs ~ Display background jobs

```
ketana_sindhu@KetaZone:~$ jobs
ketana_sindhu@KetaZone:~$
```

2. bg ~ Resume a job in the background

```
ketana_sindhu@KetaZone:~$ bg %1
-bash: bg: %1: no such job
ketana_sindhu@KetaZone:~$
```

3.fg ~ Bring a job to the foreground

```
ketana_sindhu@KetaZone:~$ fg %1
-bash: fg: %1: no such job
ketana_sindhu@KetaZone:~$
```

4.& - Run a process in the background

```
ketana_sindhu@KetaZone:~$ ./FirstFile &
[1] 801
ketana_sindhu@KetaZone:~$ Hello Linux
```

8.FILE PERMISSIONS AND LINKS

1. chmod ~ Change file permissions

```
ketana_sindhu@KetaZone:~$ chmode +x script.sh
```

2. chmod 755 ~ Set rad, write and execute permissions

```
ketana_sindhu@KetaZone:~$ chmod 755
```

3.ln ~ create a hard link

```
ketana_sindhu@KetaZone:~$ ln FirstFile.c HELLO.c
ketana_sindhu@KetaZone:~$
```

4.ln -s ~ create a symbolic link

```
ketana_sindhu@KetaZone:~$ ln -s FirstFile shortcut
ketana_sindhu@KetaZone:~$
```

9. DISK AND MEMORY INFORMATION

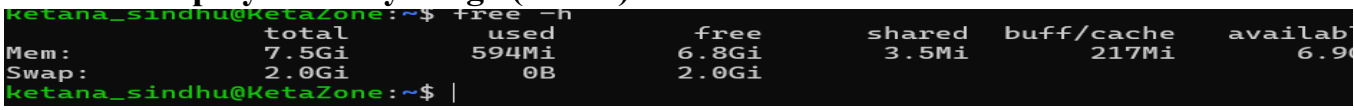
1. **df -h ~ display disk space usage**

```
ketana_sindhu@KetaZone:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G     0  3.8G   0% /usr/lib/modules/
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers         476G  141G  335G  30% /usr/lib/wsl/drive
/dev/sdd       1007G  1.7G  954G   1% /
none            3.8G   36K  3.8G   1% /mnt/wslg
none            3.8G     0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G   1% /init
none            3.8G  548K  3.8G   1% /run
none            3.8G     0  3.8G   0% /run/lock
none            3.8G     0  3.8G   0% /run/shm
none            3.8G   80K  3.8G   1% /mnt/wslg/version
none            3.8G   80K  3.8G   1% /mnt/wslg/doc
C:\             476G  141G  335G  30% /mnt/c
tmpfs           3.8G     0  3.8G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/
none            1.0M     0  1.0M   0% /run/credentials/
none            1.0M     0  1.0M   0% /run/credentials/
tmpfs           766M   12K  766M   1% /run/user/1000
ketana_sindhu@KetaZone:~$
```

2. **du -sh ~ display directory size**

```
ketana_sindhu@KetaZone:~$ du -sh
352K    .
ketana_sindhu@KetaZone:~$
```

3. free -h ~ display memory usage (Linux)

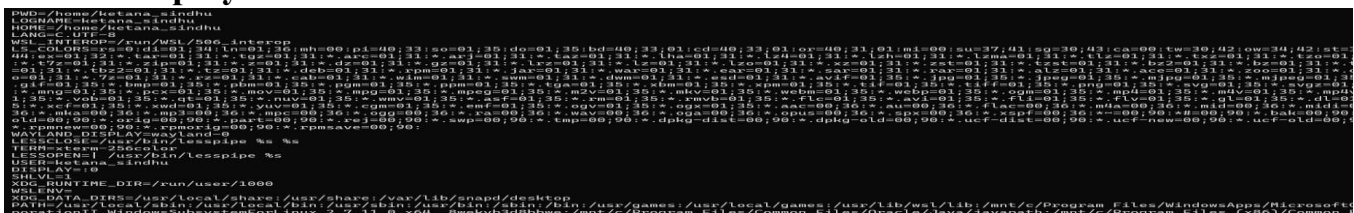


10. USER AND ENVIRONMENT MANAGEMENT

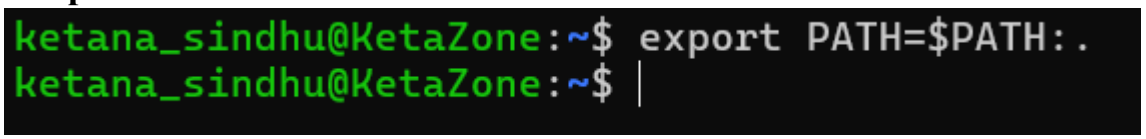
1.id ~ display user and group IDs



2.env ~ display environment variables



3.export ~ Set an environment variable



4. alias ~ Create a command shortcut

